

# Mechanical and Geometrical Properties Assessment of Thermoplastic Materials for Biomedical Application

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**Abstract** Clear thermoplastic aligners are nowadays widely used in orthodontics for the correction of malocclusion or teeth misalignment defects. The treatment is virtually designed with a planning software that allows for a definition of a sequence of little movement steps from the initial tooth position to the final desired one. Every single step is transformed into a physical device, the aligner, by the use of a 3D printed model on which a thin foil of plastic material is thermoformed. Manufactured aligners could have inherent limitations such as dimensional instability, low strength, and poor wear resistance. These issues could be associated with material characteristics and/or with the manufacturing processes. The present work aims at the characterization of the manufactured orthodontic devices. Firstly, mechanical properties of different materials have been assessed through a set of tensile tests under different experimental conditions. The tests have the purpose of analyzing the effect that the forming process and the normal use of the aligner may have on mechanical properties of the material. The manufacturing process could also introduce unexpected limitations in the resulting aligners. This would be a critical element to control in order to establish resulting forces on teeth. Several studies show that resulting forces could be greatly influenced by the aligner thickness. A method to easily measure the actual thickness of the manufactured aligner is proposed. The analysis of a number of real cases shows as the thickness is far to be uniform and could vary strongly along the surface of the tooth.

**Keywords:** 3D Human Modeling; Virtual Design; Clear Aligner; Thermoforming Process; Mechanical Properties Assessment; Optical 3D Scanner.

## 1 Introduction

Orthodontic treatment with clear aligners is becoming every day more popular as orthodontic approach, also thanks to the diffusion of the CAD/CAM technologies in biomedical applications. Compared with traditional fixed appliances with metallic braces, clear aligners are aesthetically appealing and more comfortable to wear. These treatments are preferred by adult patients, who don't want to treat their malocclusions in a traditional way. However, sometimes it is difficult to resolve complicated cases such as severe crowding, deep bites or round teeth rotations with aligners. Often there is the need of case refinement or even to turn back to braces due to the low efficacy and predictability of aligners [1-2].

The treatment process is based on the use of a digital impression, which could be obtained with a 3D scan of a physical impression of the patient mouth or by a direct scan of the desired geometry with an intra-oral scanner. The treatment is virtually designed with a planning software that allows for the definition of a sequence of little movements in order to move the tooth from the initial position to the final desired one. Every single step is transformed into a physical device, the aligner, by the use of a 3D printed model on which a thin foil of plastic material is thermoformed. However, manufactured aligners could have inherent limitations such as dimensional instability, low strength, and poor wear resistance. These issues could be associated with materials characteristics and/or with the manufacturing processes.

Mechanical performances of aligner's thermoplastic material play a critical role in developing continuous orthodontic forces able to yield the desired results. The orthodontic thermoplastic materials should have particular characteristics including transparency, lower hardness, better elasticity and resilience, and resistance to aging.

The present work aims at the characterization of the manufactured orthodontic devices. Firstly, mechanical properties of three different materials commonly used in orthodontic practice has been assessed through a set of tensile tests carried out under different conditions. The tests have the purpose of analyzing the effect that the forming process, and subsequently the normal use of the aligner, may have on properties of the material. For this reason, the samples were subjected to an aging treatment in a particular solution that reproduces the biochemical behavior of human saliva.

On the other way, also the manufacturing process could introduce unexpected limitations in the resulting aligner [3-4]. This would be a critical element to control within the aligner in order to explore resulting forces on teeth. Moreover, force application in the tooth-aligner system is not a fully understood mechanism. Several studies show that it could be greatly influenced by the aligner thickness [5-7]. A method to easily measure the actual thickness of the manufactured aligner is proposed. The method is based on the use of an optical scanner to acquire the physical model used for thermoforming the device with and without the aligner.

The difference between the two acquisitions depends only on the thickness of the aligner. The analysis of a number of real cases shows as the thickness is far to be uniform and could vary strongly along the surface of the tooth.

Results obtained in the present work have allowed to characterize the aligner in terms of mechanical and geometrical properties with high level of confidence. This allows for a more accurate definition of the problem and a better design of the treatment planning. Results could be also used to realize a reliable finite element method (FEM) model to use for treatment optimization purposes [8-9].

## **2 Materials and methods**

### ***2.1 Mechanical properties***

Aligners in the oral cavity are subjected to aggressive environment that could lead to a high degradation of their mechanical properties with negative influence on treatment efficacy. For this reason, tensile tests were extended to different laboratory conditions to simulate the actual operating conditions. In particular, an aging stage was designed to reproduce the influence of the saliva corrosion on the aligner. An artificial mixture was used to reproduces the biochemical behavior of human saliva [10-11].

#### ***2.1.1 Tensile test***

The test campaign was designed on the EN ISO 527-1:2012 [12] technical indications that establishes general conditions and procedures for tensile tests on plastic materials.

For each of the three materials, tests in original conditions (as from supplier), after the thermoforming stage and after thermoforming and aging stage, were carried out. All materials were supplied in circular plates (diameter 125 mm) with thickness of 0.75 and 1 mm. Both shape and size of the specimens were chosen on this thickness value as suggested by normative.

The thermoforming process was performed under manufacturer's recommendations. The thermoformed specimens were obtained by using the same technological process adopted to manufacture the aligner. A 3D printed disk of 80 mm diameter and 12 mm height was used as model form for the thermoforming machine (Ministar S from Scheu Dental®). The parameter used was 70°C for the infrared

lamp and 4 bar pressure for 30 seconds. The flat surface on the top of the disk provides a flat area of thermoformed material, which can be used to obtain the desired specimens. Final specimens were then obtained by high-pressure waterjet cutting and manually refined. Each specimen was then measured with a micrometer (CLM1-15QM, Mitutoyo Co.) to establish the cross section area and preliminary weighted to evaluate the water quantity absorbed after the aging stage (Table 2). Some samples were subjected to the aging treatment. The aging stage consisted in a bath in the prepared compound (Table 1) for 7 days at an environmental temperature of 37°C. This time corresponds to half the normal use time of each aligner during the actual treatment. It is justified because the water absorption from plastic materials mainly occurs in the first 72-168 hours [13]. At the end of the aging stage, the material was placed in a special environment (silica gel bell) in order to reach the equilibrium condition and its weight variation was evaluated to establish the absorbed rate of water.

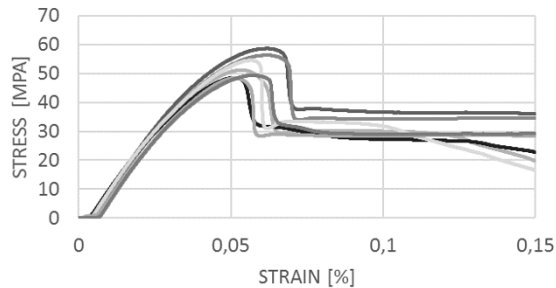
The tensile tests were performed by the Instron Universal Testing Machine 5500R at a temperature of 23°C. A total of 10 specimens for each material were tested under each condition (figure 1). Two different test campaigns were designed to determine the elastic modulus and the tensile yield stress. The testing speed was defined as 0.1 mm/min to obtain data for the elastic modulus and 5 mm/min for the stress-strain curves as suggested by the ISO527-1. Mean and standard deviation values were determined for each item. Comparisons among materials under each test condition were finally performed.

**Table 1.** Chemical composition of artificial saliva pH=6.5.

| <i>Compound</i>                                      | <i>Content [g/l]</i> |
|------------------------------------------------------|----------------------|
| NaCl                                                 | 0.6                  |
| KCL                                                  | 0.72                 |
| CaCl <sub>2</sub> ·2H <sub>2</sub> O                 | 0.22                 |
| KH <sub>2</sub> PO <sub>4</sub>                      | 0.68                 |
| Na <sub>2</sub> HPO <sub>4</sub> ·12H <sub>2</sub> O | 0.856                |
| KSCN                                                 | 0.06                 |
| NaHCO <sub>3</sub>                                   | 1.5                  |
| Citric Acid                                          | 0.03                 |

**Table 2.** Weight variation of material specimen for water absorption.

| <i>Material</i> | <i>Before aging (g)</i> | <i>After aging (g)</i> | <i>Variation %</i> |
|-----------------|-------------------------|------------------------|--------------------|
| Mat 1           | 0.1334                  | 0.1339                 | 0.375%             |
| Mat 2           | 0.1255                  | 0.1261                 | 0.438%             |
| Mat 3           | 0.0974                  | 0.0981                 | 0.719%             |



**Fig. 1.** Sample of the stress-strain curve obtained for a set of specimens of the material “*Mat 1*” in original supplier condition.

### 3.1 Geometrical properties

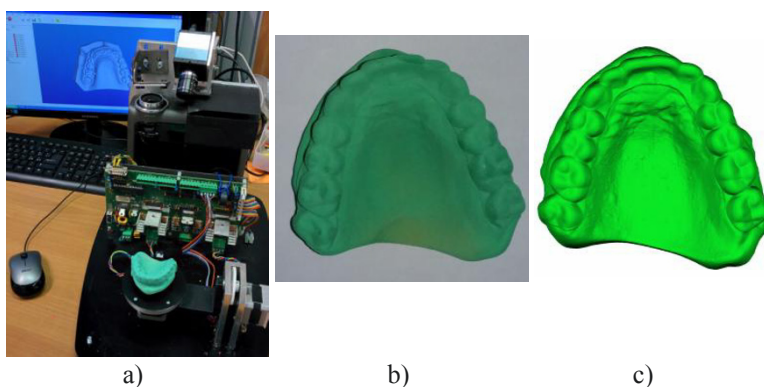
The manufacturing process could introduce unexpected variations of the geometry of the resulting aligner. The measurement of the actual aligner’s shape would be a critical element in order to assess the force system delivered by the appliance. In this work, a method to easily measure the actual thickness of the manufactured aligner is proposed. The method is based on the use of an optical scanner to acquire the physical model used to thermoform the device with and without the aligner worn on it. The difference between the two acquisitions depends only on the thickness of the aligner.

#### 3.1.1 3D optical measurement

In this work, the focus was on the measurement of the thickness of the aligners to assess its value along the teeth surfaces. At this purpose, several aligners were manufactured with the same technological approach used for the usual aligner production. Reference models of an upper and a lower human arch were 3D printed by a Stratasys Eden500V polyjet machine in high quality printing setup (slice thickness of 16 $\mu$ m, accuracy of 200 $\mu$ m). The material used was the VeroDent MED670 as construction material and SUP705 as supporting material. The supporting material was finally removed with high-pressure water jet in order to obtain the final models. The models were used to thermoform the aligners samples, with the same parameters used for the thermoformed samples of the tensile test (§ 2.2.1). Finally, the aligners were manually cut and refined with specific milling tools.

An optical 3D scanner, specifically developed for dental models [14-15], was used to measure the real shape of the aligner.

The acquisition methodology is based on an active stereo vision approach, which uses a binary coded lighting (fringe projection) to recover 3D points of target surfaces. The hardware setup is composed of a DLP<sup>®</sup> projector (OPTOMA EX330e, resolution XGA 1024x768 pixels) and an 8-bit monochrome charge-coupled device (CCD) digital camera (The Imaging Source<sup>®</sup> DMK 41BF02, resolution 1280x960 pixels) equipped with a lens having focal length of 16mm (PENTAX, C31634KP 2/3 C-mount). Camera and projector are used as active devices of a stereo triangulation process, and the stereo rig has been configured for a working distance of 300mm and a working volume of 100×80×80mm, with a resulting lateral resolution of 0.1mm and accuracy of 10 $\mu$ m. A calibration procedure is required to calculate the intrinsic and extrinsic parameters of the optical devices, with respect to an absolute reference system. The optical devices have been integrated with a double motorized turntable (figure 2a). The acquisition system allows for a complete automatic measurement of dental model arches.



**Fig. 2.** Model placed on the acquisition plate of the 3D scanner (a), detail of the 3D printed model with aligner (b) and the result of the acquisition (c).

The methodology developed for the measurement of the aligner's geometry consists in the acquisition of the model used for the thermoforming process with and without the aligners fitted on it. The model and the aligner surfaces were uniformly covered with an optic spray (Occlu Plus Hager & Werken) in order to have a uniform non transparent color (figure 2b). The software Raindrop Geomagic Qualify<sup>®</sup> was used to compare the results. The two acquisition were aligned in a common reference system by a best fit matching algorithm. Differences were evaluated in terms of 3D distances between corresponding points of the surfaces and 2D distances between some main slices of the two digital models.

4 Results and discussion

The mechanical properties of thermoplastic materials used for the production of clear aligners may greatly vary from the nominal values for what concerns elastic modulus and tensile yield stress. The variation can be influenced by several factors, such as manufacturing technologies and use conditions. This study focused on the assessment of material’s mechanical properties by reproducing some of the actual use conditions, in order to establish the values of changes rate. Moreover, an assessment of the geometrical shape of the manufactured aligners has been carried out.

4.1 Tensile tests

Tables 3 and 4 and the graphical plots in the figure 3 show the results for elastic moduli and tensile yield stress for materials analyzed under the different conditions.

Table 3. Results from tensile test for elastic modulus *E*.

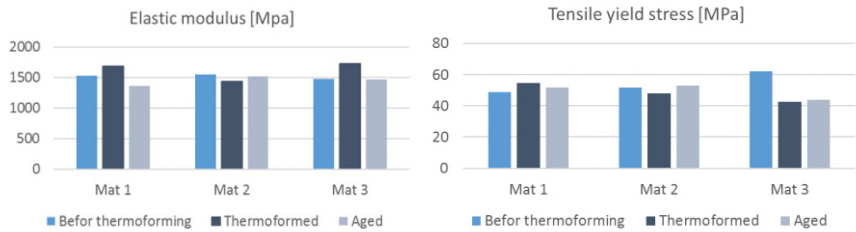
| <i>Material</i> | <i>Condition</i>              | <i>E</i> [MPa] | <i>σ</i> |
|-----------------|-------------------------------|----------------|----------|
| <i>Mat 1</i>    | <i>Supplier specification</i> | 2200           | -        |
|                 | <i>Before thermoforming</i>   | 1531           | 41       |
|                 | <i>Thermoformed</i>           | 1693           | 51       |
|                 | <i>Aged</i>                   | 1368           | 35       |
| <i>Mat 2</i>    | <i>Supplier specification</i> | 2050           | -        |
|                 | <i>Before thermoforming</i>   | 1556           | 48       |
|                 | <i>Thermoformed</i>           | 1447           | 42       |
|                 | <i>Aged</i>                   | 1519           | 62       |
| <i>Mat 3</i>    | <i>Supplier specification</i> | -              | -        |
|                 | <i>Before thermoforming</i>   | 1478           | 88       |
|                 | <i>Thermoformed</i>           | 1730           | 77       |
|                 | <i>Aged</i>                   | 1466           | 72       |

Results show how the thermoforming process generally does not lead to a great decrease in elastic properties of the thermoplastic materials, except for one of the tested materials (Mat 3), which shows a 30% decrease in tensile yield stress after thermoforming. For the other materials, a maximum properties variation of 15%, with respect to the supplier specifications, can be observed. A general reduction of the elastic modulus has been measured with respect to values declared in the technical sheet (except for the Mat 3 for which no technical data were found), while

very similar values were found for tensile yield stress. In some cases, a little increase in values has been obtained after the thermoforming process.

**Table 4.** Results from tensile test for tensile yield stress.

| <i>Material</i> | <i>Condition</i>              | <i>Tensile yield stress [MPa]</i> | <i><math>\sigma</math></i> |
|-----------------|-------------------------------|-----------------------------------|----------------------------|
| <i>Mat 1</i>    | <i>Supplier specification</i> | 53                                | -                          |
|                 | <i>Before thermoforming</i>   | 49,29                             | 0,45                       |
|                 | <i>Thermoformed</i>           | 53,52                             | 4,84                       |
|                 | <i>Aged</i>                   | 49,49                             | 1,76                       |
| <i>Mat 2</i>    | <i>Supplier specification</i> | 50                                | -                          |
|                 | <i>Before thermoforming</i>   | 52,10                             | 1,49                       |
|                 | <i>Thermoformed</i>           | 48,75                             | 2,57                       |
|                 | <i>Aged</i>                   | 50,62                             | 2,88                       |
| <i>Mat 3</i>    | <i>Supplier specification</i> | -                                 | -                          |
|                 | <i>Before thermoforming</i>   | 62,37                             | 0,90                       |
|                 | <i>Thermoformed</i>           | 41,92                             | 2,94                       |
|                 | <i>Aged</i>                   | 44,61                             | 1,82                       |



**Fig. 3.** Distribution of elastic modulus and tensile yield stress in the tested conditions.

Intraoral environment could also lead to changes in material mechanical properties. The main influence is from the rate of water absorbed and the aggressive characteristic of the human saliva. These effects have been taken into account by the tensile tests on aged material. The results, in this case, show that the elastic modulus generally decreases with respect to both the supplier specifications and the simply thermoformed material. It is also worth noting that water absorption could affect the sizes of the aligner and have an influence on the fit on the patient arch, with potential loss of the effectiveness by unpredictable orthodontic force on the teeth.



## 4.2 Geometrical measurement

The correct use of thermoplastic aligners in orthodontic practice requires the knowledge of the real geometry of the actual aligner, particularly about its thickness along surfaces. The change of the shape is surely influenced by the hygroscopic property of the material, which needs to be assessed. Indeed, the standard production technology used for the fabrication of the aligners has the main influences on the resulting shape of the object. In this work, a method to measure the resulting size of the aligners was proposed in order to relate the real shape object with the technological approach adopted to produce the aligner.

The acquisitions show that the thickness of the artefacts is far away from being uniform and can vary significantly along its surface. Figure 4 shows the measured shapes of the aligners.

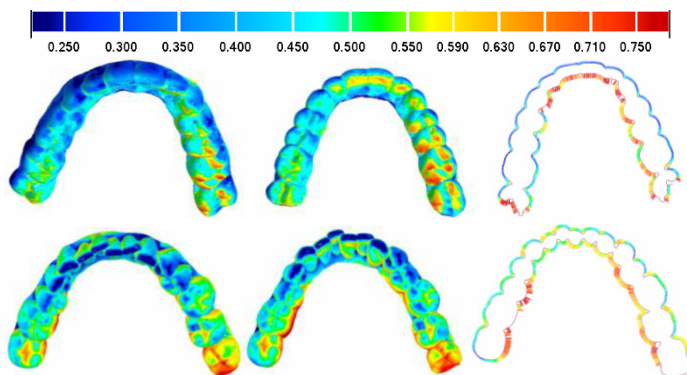


Fig. 4. 3D compare map and 2D measurement on a slice of an upper and a lower arch.

## 5 Conclusion

Mechanical properties of the materials used for the production of clear aligners are of utmost importance to evaluate the effectiveness of an orthodontic treatment with these devices. Technical data supplied along with materials cannot always be used as a reference, but need to be assessed taking into account the different conditions in which they are used. Results obtained in the present work suggest that the magnitude of the effects of the different conditions on the mechanical properties of thermoplastic materials can differ between different materials. Therefore, materials for orthodontic appliances should be selected after a detailed characterization of their mechanical properties in the simulated intraoral environment. On the other way, the force delivery process in the tooth-aligner system is not a fully understood mechanism. Several studies show that it could be greatly influenced by the aligner thickness. In this work, a methodology to measure geometrical proper-

ties of clear aligners is proposed. The results obtained for real cases show as the thickness is far to be uniform and could vary strongly along the surface of the tooth. Future works will be dedicated on the validation of the proposed method by its application for optimize real orthodontic cases.

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